The adjoint of the trace operator:
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We know the standard trace operator acts on linear transformations of H onto itself. That is, we have a vector or Hilbert space, H, and linear operators on the vectors in H. This space of linear operators on H is called $L(H)$.

The trace operation is a *super operator* in Watrous' terms in that it is linear but acts on elements of $L(H)$ instead of H. And it maps from $L(H)$ into the complex numbers which can be viewed as a one-dimensional complex vector space, $\mathbb{C}$.

That is,

$$tr : L(H) \to \mathbb{C} \tag{1}$$
$$X \mapsto y = trX. \tag{2}$$

We know this map is linear, which can be seen by expanding the trace in coordinates for some ONB.

This means there exists a unique *reverse* map from $\mathbb{C}$ to H, that we call $tr^\dagger$,

$$tr^\dagger : \mathbb{C} \to L(H), \tag{3}$$

such that inner products are preserved:

$$(tr^\dagger y, X)_{L(H)} = (y, trX)_{\mathbb{C}} \tag{4}$$

This is shown in Figure 1 for a general operator, A, from K to $\mathbb{C}$ but holds for any target vector space with an inner product.

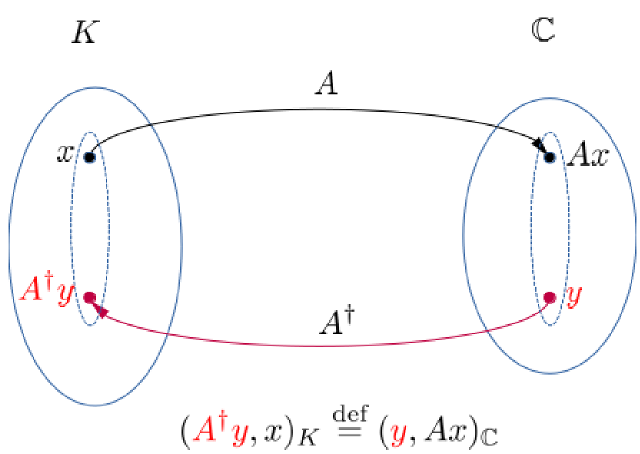

Figure 1: Diagram showing compatibility of inner products for the adjoint of a general operator $A : K \to \mathbb{C}$. In the most general case we can replace $\mathbb{C}$ with any other vector space N and its inner product, $(\cdot, \cdot)_N$.

As we went over in the talk and the notes it is easy to show such a reverse map exists, is unique and is linear.

To actually construct it for a general operator A we use the expansion in coefficients,

$$A^\dagger y = \sum_{i=1}^{\dim K} c_i e_i, \tag{5}$$

where e_i are an ONB for K and the c_i are the expansion coefficients to be solved for.

We showed that we can evaluate the coefficients in terms of A and y, giving

$$c_i = (Ae_i, y)_K, \tag{6}$$

so we can explicitly construct the action of the adjoint as,

$$A^\dagger y = \sum_{i=1}^{\dim K} (Ae_i, y)_K \; e_i. \tag{7}$$

In our case of the trace, the basis vectors for $L(H)$ are the basis matrices E_{ab}, with a 1 in the (a,b) position and zeros everywhere else. Since the codomain is $\mathbb{C}$ here, the inner product in $\mathbb{C}$ is just $(x,y)_{\mathbb{C}} = x^* y$, where superscript $*$ is the usual complex conjugate.

So for any complex number, y, we can expand the adjoint of the trace acting on it as,

$$\begin{aligned}
tr^\dagger y &= \sum_{a,b} (tr \; E_{ab}, y)_{\mathbb{C}} E_{ab} \\
&= \sum_{ab} (\delta_{ab}, y) E_{ab} \\
&= \sum_{ab} \delta_{ab} \; y \cdot E_{ab} \\
&= y \sum_{a} E_{aa} \\
&= y \cdot \mathbb{I}_{L(H)}
\end{aligned}$$

So we have the adjoint of the trace operator acts on a complex scalar by multiplying by the identity matrix for the original vector space:

$$tr^\dagger y = y \cdot \mathbb{I}_{L(H)}. \tag{8}$$

As a check we can substitute this into the defining inner product compatibility in (4),

$$(tr^\dagger y, X)_{L(H)} = (y, trX)_{\mathcal{C}}. \tag{9}$$

This gives us,

$$\begin{aligned}
lhs &= (y \cdot \mathbb{I}_{L(H)}, X)_{L(H)} && (10)\\
&= tr\left((y \cdot \mathbb{I}_{L(H)})^\dagger X\right) && (11)\\
&= y^* \; tr(X), && (12)
\end{aligned}$$

$$\begin{aligned}
rhs &= (y, trX)_{\mathbb{C}} && (13)\\
&= y^* \; tr(X), && (14)
\end{aligned}$$

and they are equal as expected.